SPIDER — System for Pedestrian Intent Detection and Emergency Response

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1. Introduction

Problem: ADAS braking is reactive — it starts only once a pedestrian is already in the path. At 40 km/h a car covers about 11 m every second.

Idea: predict whether a tracked pedestrian will cross (Crossing / Not-Crossing) from 15 frames of motion, and link that decision to braking.

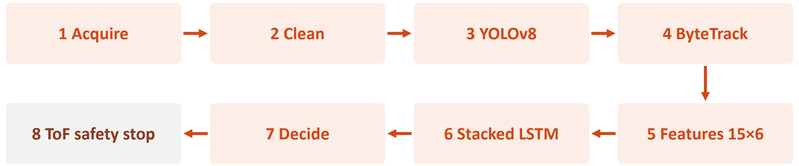
Objectives: restore and calibrate a Duckiebot (Jetson Nano); establish ROS + rosbridge communication; deploy the YOLOv8 + ByteTrack + stacked-LSTM pipeline; add front time-of-flight ranging.

Scope: controlled indoor tests with a safety operator; binary intent; off-the-shelf sensors; braking through drive and e-stop commands (no custom brake actuator).

2. Methodology

Pipeline: clean (confidence filter, Laplacian blur test, gap interpolation) → YOLOv8 → ByteTrack → features (xc, yc, w, h, dx, dy) over T = 15 → stacked LSTM (64 → 32) → crossing probability, threshold 0.5.

Decision: in-path + distance + probability → Brake-Engaged; an independent ToF safety stop always wins. Inference runs on a workstation linked by rosbridge.



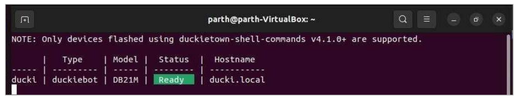
Per-frame flow redrawn from the report's algorithm (§3.3); step 8 runs alongside step 7.

3. Implementation

Platform: Duckiebot DB21M, Jetson Nano 2 GB, HAT v3.1, IMX219 camera + raised USB webcam, front ToF (0.05–1.2 m).

Bring-up: re-imaged; ROS Noetic in Docker; intrinsic + extrinsic camera calibration; kinematic trim −0.0625 over 2 m.





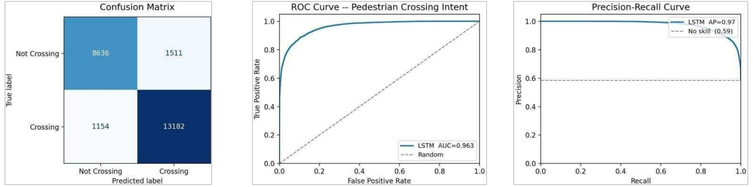
Left: experimental setup (Fig. 4.2). Top: robot reported Ready (Fig. 4.1, cropped).

rosbridge topics: image + range → workstation; drive (v, ω) + e-stop → robot.

4. Results & Testing

JAAD test split — dataset result (24,483 samples): accuracy 89.11% (baseline 58.5%), AUC-ROC 0.963, AP 0.974, crossing recall 0.920, miss rate 8.05%.

Qualitative: Indian-road footage shows Crossing / Not-Crossing tags and Brake-Engaged decisions; robot checks are reported without numbers; latency and on-device results are not yet measured.



Confusion matrix, ROC and precision–recall on the JAAD test split (Appendix C).

5. Innovation

Combination, not a single component: pre-model data cleaning; rangefinder fused with motion intent; obstacle-agnostic ToF safety net beneath the learned decision; detection-first hazard gate biased toward stopping; manual-drive override.

Aspect	Typical existing work	SPIDER
Inputs	Clean tracks assumed	Explicit cleaning stage
Distance	Appearance-only / none	ToF front range + hard stop
Deployment	Offline benchmark	ROS robot; inference off-board
Braking	Alert / not addressed	Two-layer + safety net
Override	Rarely addressed	Manual-drive e-stop

Qualitative comparison (Table 6.1); no numeric benchmark against prior models was reported.

6. Application & Outcome

Relevance: anticipatory pedestrian braking for dense, unconstrained Indian traffic, on a low-cost testbed meant to port to a larger vehicle.

Paper: Real-Time Pedestrian Crossing Intention Prediction Using Vision-Based ADAS



Fig. 5.2: Crossing (red) and Not-Crossing (green) tags with a Brake-Engaged decision on Indian-road footage.

Qualitative; footage processed off-board.

7. Conclusion & Future Scope

Achieved: a crossing-intent model validated on JAAD (89.11% accuracy, 0.963 AUC, 0.974 AP); a restored, calibrated Duckiebot with ROS and rosbridge; a two-layer braking design with a ToF safety net and manual override; qualitative Crossing / Brake-Engaged results on Indian-road footage.

Still open: on-device (Jetson) real-time inference, quantified braking trials and measured reaction time.

Future scope: ONNX/TensorRT deployment on the Jetson with latency and FPS measurement; radar/LiDAR for fog, rain and night; graded intent states; group-aware models for dense crowds; signal-aware braking; a larger multi-sensor dataset; migration to a scaled vehicle.